

- Naïve Bayes classifiers
- Decision trees
- Support vector machine

We will use Google Colab, which is an open source development tool

published by Google and widely accepted over the Internet and by the ML fraternity, for the initial analysis. It has built-in GitHub and Google Drive integration. It's a free Jupyter

notebook with GPU support, and has a varied range of features that can integrate with Kaggle.

A very apt diagrammatic differentiation between the two types of supervised learning techniques can be seen in Figure 2b.

Regression technique

In case of supervised learning, the major concept is to baseline the prediction on the basis of labelled data and the target variable, which is of continuous nature. Supervised learning is differentiated between classification and regression. In the case of categorical data, we use different classification techniques, while in this example, we will focus on multiple linear regression. This is defined as a statistical technique used to predict the outcome of a response variable through several explanatory variables and model the relationships between them.

As part of the detailed implementation phase, we will start with data preparation and cleansing to remove biases and errors. We will use Python techniques to remove the null values in the data set. The Python pandas dataframe library has methods to help data cleansing as shown below. This method will remove the null values in the data set:

```
# Removing the missing value rows in the dataset
dataset = dataset.dropna(axis=0, subset=['Year', 'Publisher'])
dataset.isnull().values.any()
```

We will now derive the dependent and independent variables out of different attributes in the columns. Our prime goal is to review the relationship between independent and dependent variables, denoted by x and y , respectively. This can be achieved by using the heat map technique to derive the correlation between different features, and

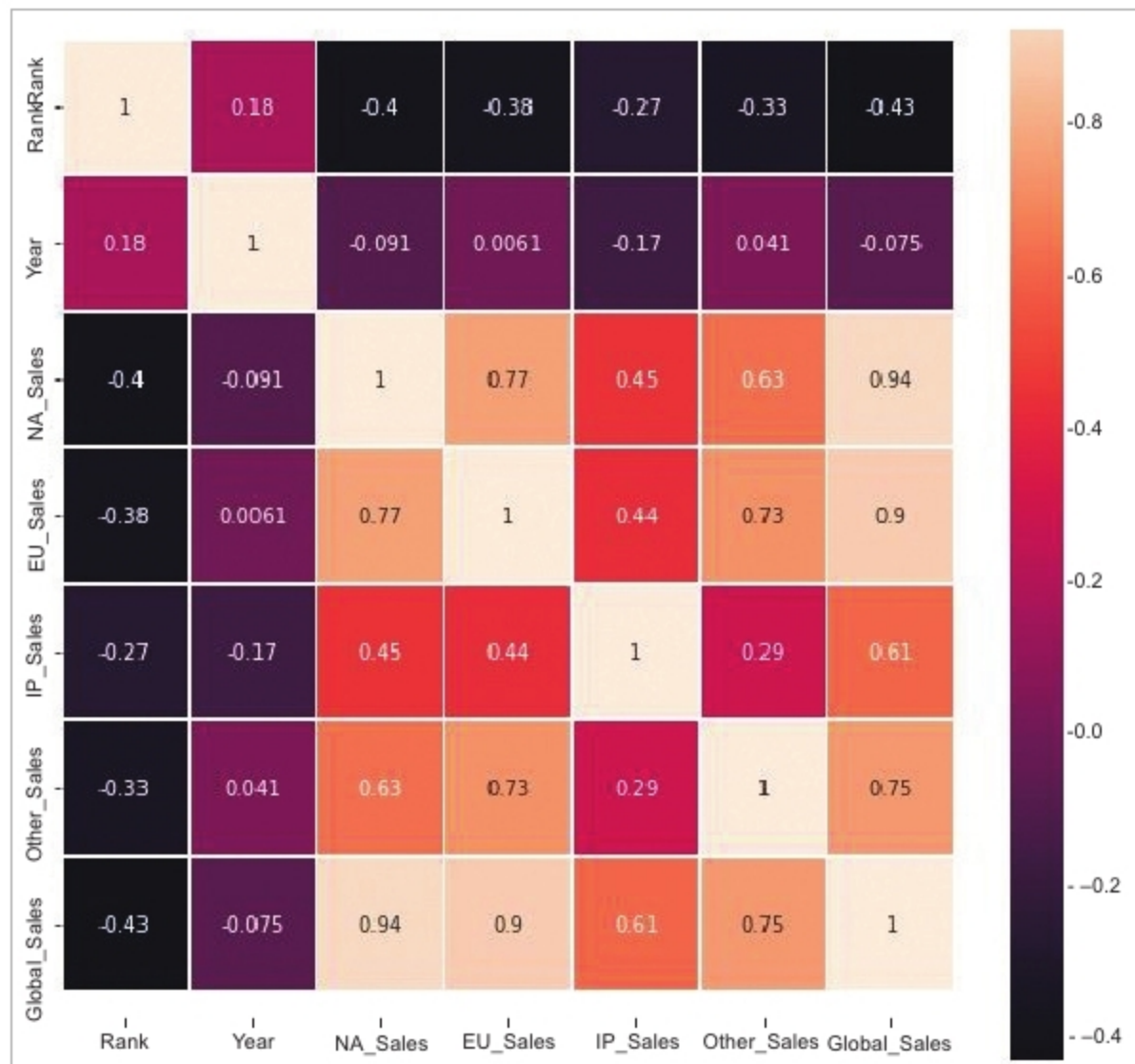


Figure 4: Heat map to represent 2D visualisation of correlation between attributes or features

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
from google.colab import files

uploaded = files.upload()

## Importing the dataset
dataset = pd.read_csv('vgsales.csv')
dataset.info()
```

Choose Files vgsales.csv

- vgsales.csv(application/vnd.ms-excel) - 1433 bytes, last modified: 11/6/2020 - 100% done

Saving vgsales.csv to vgsales (1).csv

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19 entries, 0 to 18
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Rank        18 non-null    float64
```

Figure 5: Colab screenshot for uploading data files